
Is the society you live in really stable?

Summary

Since entering the 21st century, the competition in the current society has become increasingly fierce. The world has seen an unprecedented number of economic development tasks and contradictory risk challenges. Among them, social stability risks involve the part of main structure and have the closest relationship with human beings as a whole, while directly affecting the security of state power. We have constructed a comprehensive evaluation system of social stability and an early warning model of social instability risk, hoping to provide constructive advice in the social stability work of each country.

For question 1, for sub-question 1, based on scientific, systematic, dynamic and operable, we construct 8 dimensions and subdivide 17 specific indicators, and calculate their final scores by **Entropy-Weighted-TOPISIS** model. **For sub-question 2**, the **Spearman correlation coefficient** is used, and it is found that GDP per capita has a strong positive correlation with hospital beds per capita, R&D expenditure, and high-tech products... **For sub-question 3**, on the basis of sub-question 2, the causal relationship between indicators with strong correlation is further explored, and it is found that GDP per capita is the causes...

For question 2, considering the basic requirements as well as the coordination of development, we build an early warning model consisting of the **integrated level**, **coordination level** and **basic requirements**, where the integrated level is measured by the Entropy-Weighted-TOPISIS score in question 1. What's more, the coordinated development level is measured by the **Coupling Coordination Degree**, and the basic needs level is represented by freshwater per capita, GDP per capita, and unemployment rate. Then the early warning level is then divided into five levels based on the assessment of these five indicators.

For question 3, we choose **Cambodia** which had a color revolution in 2014. We use the 2014 data for analysis and find that the GDP per capita, coordination level, and comprehensive level do not meet the requirements we set and give it an early warning level of near-severe. The failure of the color revolution is because mainly because it did not have the problem of disputes over religious beliefs, had relatively low dependency, and had more freshwater resources per capita. Combining various domestic and international factors, we give the predicted result that no color revolution will break out.

For question 4, we choose **Myanmar** which had a color revolution in early 2021. Since there is a certain lag in the data, we use the data from 2020 for our analysis and find that the GDP per capita, dependency ratio, comprehensive level, and coordination level do not meet the requirements we set, and give it an early warning level of serious risk. The main reasons for the success of the color revolution are: its high unemployment rate, low GDP per capita, and low R&D spending.

For question 5, based on the results of our theoretical and empirical studies, several construction ideas are proposed to address the existing problems, mainly the following.

- Focus on **economic development** and increase the material wealth of society
- Make good use of the **demographic dividend** and gain capital and technology
- Increase stable **employment** and guarantee people's basic livelihood
- Improve the **competitiveness of science and technology** and build a strong country in science and technology
- Establish a **social safety valve mechanism**, including a "communication mechanism at the upper and lower levels" and a "third-party mediation mechanism"

Keywords: Social stability assessment, Social stability risk warning, Entropy-TOPISIS model, Coupling coordination degree

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1 Introduction

1.1 Problem Background

Since entering the twenty-first century, in the increasingly competitive and crisis-ridden society of the present. The world has seen an unprecedented number of economic development tasks and contradictory risk challenges. Among them, social stability risks involve the largest number of actors and have the closest relationship with human beings as a whole, while directly affecting the security of state power. These are also highly valued by leaders and politicians at present[1].

Social stability risks are influenced by multiple factors, including the level of local economic development, the spiritual life of the population and the international situation of the country, and the factors can also interact with each other[2]. Once one of these factors fluctuates greatly, it will affect other factors and even the stability of the whole society, which will lead to social unrest.

Therefore, in the face of social stability, which is of great importance, we must prevent it before it happens and intervene before it generates great social unrest. This requires the establishment of a social stability risk assessment model to determine the intrinsic connection between social stability risks and the current country, including economic, political, and cultural aspects, by studying past major events affecting social stability, and intervening in a timely manner to eliminate the risks[3]. We can learn from history and know ourselves.

However, social stability risk assessment models are extremely life-cycle, subject-specific, and also need to ensure broad coverage of content. Since the 17th century, models of social stability risk assessment have emerged, with politicians and scholars hoping to find a general solution for society, but each of the resulting models inevitably has a distinctive contemporary flavor. With the development of economy and society, the social psychology of the public has become more complex and the uncertainty of social stability has greatly increased.

Based on this, our team constructed a social stability indicator system in conjunction with the current social situation as a way to conduct social stability early warning and build a beacon of social stability. We also used our model to explain two famous color revolutions and pointed out the reasons for their success or failure, and finally, we put forward our own views and suggestions for maintaining social stability.

1.2 Restatement of the Problem

➤ Question 1

For sub-question 1, representative indicators are constructed by establishing 8 major dimensions with a total of 17 specific indicators, including 4 qualitative indicators and 13 quantitative indicators, to measure various aspects of a country's stability, including resource endowment, economic situation, medical level, education level, science and technology level, personal stress, comprehensive strength, and ideology. we have established the entropy weight-TOPSIS model to analyse Forty countries during 2014 to 2020. For sub-question 2, the correlations among the 8 evaluation dimensions are calculated through the Spearman correlation coefficient, and then the correlations among the indicators are derived. For sub-question 3, some indicators with correlations have been derived through the analysis of sub-question 2, and then the causality of these correlated indicators is obtained through the analysis.

➤ Question 2

In order to establish a more scientific and reasonable early warning model of social stability, this paper analyzes the balance of the development of each dimension of a country comprehensively by constructing a coupled coordination degree model of 8 dimensions, and at the same time, sets the corresponding thresholds according to 17 specific indicators as the boundaries of evaluation, and integrates the entropy-weight-TOPSIS score, coupled coordination degree and sub-thresholds as the basis of early warning, according to the breach of the boundaries of number and combination, it is divided into serious warning, near serious warning, moderate warning, light warning, and no warning.

➤ Question 3

Cambodia was chosen as the study subject that attempted a color revolution but turned out to be a failure, and its color revolution was in 2014, so the data from 2014 was chosen for analysis. After the analysis, we conclude the reasons for the failure of its color revolution.

➤ Question 4

Myanmar was selected as the subject of a successful color revolution whose color revolution occurred in early 2021, so the data for 2020 is chosen for analysis. After analyzing the data, we conclude the reasons for the success of the color revolution.

1.3 Our Work

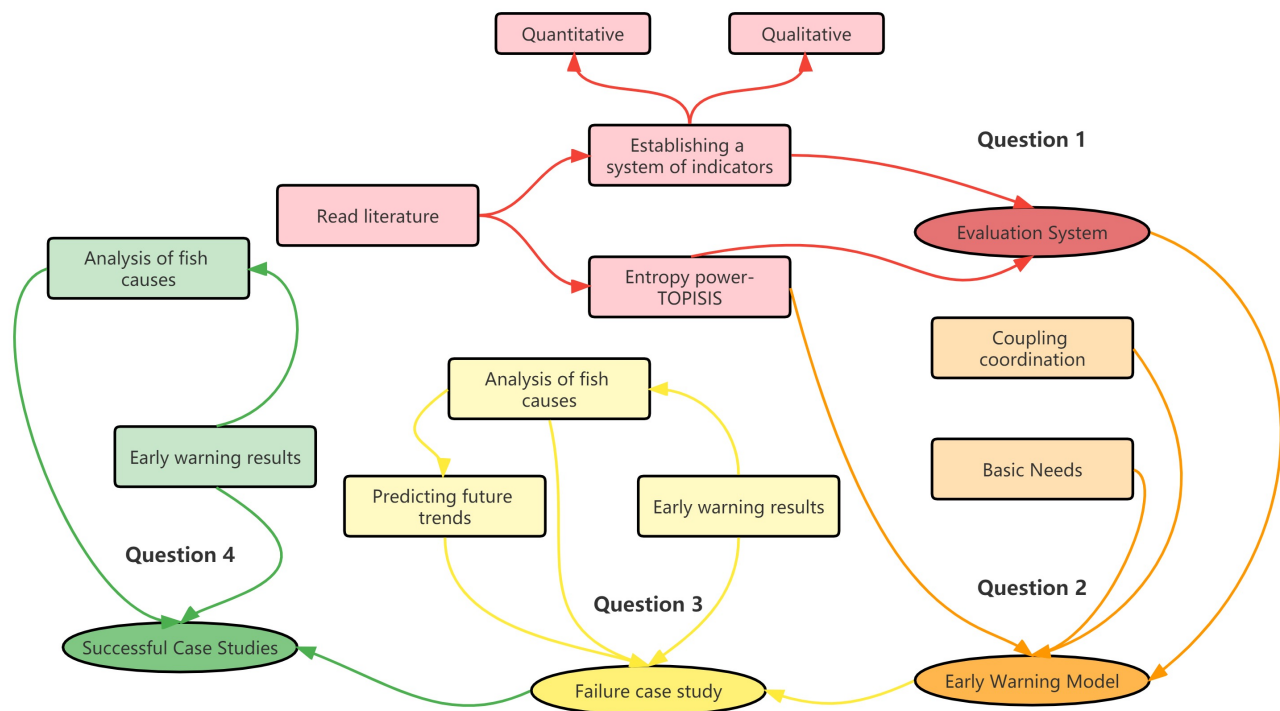


Figure1 : Work flow chart

2 Assumptions and Justifications

➤ Hypothesis 1

The outbreak of color revolutions is the result of a combination of internal and external causes.

Rationale: Although color revolutions are traditionally seen as regime change actions without violence and armed struggle by Western capitalist countries against socialist countries in the Soviet camp, it is undeniable that the internal factors of their countries also have an important influence on this. .

➤ Hypothesis 2

There is a certain internal consistency in the countries where color revolutions break out, and their patterns can be explored.

Rationale: If the selection of countries where color revolutions break out is completely random, then this study cannot be conducted at all.

➤ Hypothesis 3

The stability of societies and their crisis nature can be assessed and predicted by past as well as current data.

Rationale: Since social development has a certain inertia, it is possible in the field of sociological research to explain the quo through past patterns and to predict future trends.

➤ Hypothesis 4

Various factors have different degrees of influence on social stability.

Rationale: Since society is constantly evolving, the influence of each factor on social stability is not static, and the degree of influence evolves with the changing and developing times.

3 Notations and Glossary

3.1 Definitions

➤ Definition 1

Color revolutions are color-named, peaceful, non-violent regime change movements that occurred in the former Soviet Union and the Middle East and North Africa in the early 21st century

➤ Definition 2

Social stability perturbations: society as an open dynamic system is bound to operate with perturbations from both internal and external sources, some of which can force the social system to deviate from its steady state.

➤ Definition 3

Social stability composite index: It is the distance between a certain indicator and the positive or negative ideal value of a country calculated by objective weighting, and finally the value calculated by synthesis, which represents the comprehensive level of social stability of a country.

➤ Definition 4

People's livelihood security system. In the natural world, human society is a survival system with social nature. If this survival requirements are not effectively guaranteed, the society will lose its stability. Therefore, the core of the social stability system structure is the livelihood security issue, and all other systems are the extensions and extensions of the human survival issue.

➤ Definition 5

Risk warning level: It refers to the assessment of social unrest and regime change due to the outbreak of color revolution in a certain country within a short period of time, and is a prediction of the probability of occurrence at a certain time.

3.2 Notations

Table1 : Symbol description

Symbol	Description
a_{ij}	Raw value of an indicator for a country
x_{ij}	Standardized value of an indicator for a country
p_{ij}	The sum of the standardized value of an indicator for a country and plus 0.0001
lnm	m is the number of countries
e_j	Information entropy of an indicator
w_j	Weight of an indicator
D_i^+	Distance between a country's score and the maximum value
D_i^-	Distance between a country's score and the minimum value
s_i	Final score
AS	Entropy-weighted-TOPISIS composite score
U_i	Standardized result of the ith indicator
C	System coupling degree
T	System development degree
D	System coupling coordination degree

4 Model Establishment and Solutions

4.1 Task1

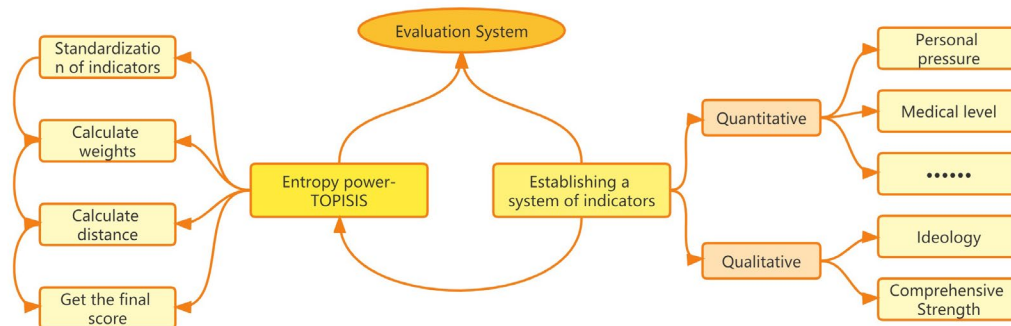


Figure2 : Work flow chart

4.1.1 Entropy-TOPSIS Model

The entropy-weighted TOPSIS method refers to first using the entropy-weighted method to assign weights to each indicator within the social stability system, then using the TOPSIS method to derive the positive and negative ideal values of each indicator, and then comparing each indicator with the positive and negative ideal values to derive the final social risk evaluation results[4]. Combining the objective assignment of entropy method and the multi-attribute decision ranking of TOPSIS method, the weights obtained by entropy method are operated in the process of constructing the weighted normalized matrix by entropy TOPSIS method, and in this way, the objective assignment method is skillfully linked with the rigorous multi-dimensional ranking to help decision makers make scientific evaluation[5]. Compared with other evaluation methods of financial performance entropy TOPSIS method has objectivity in the selection of indicators and calculation of weights, etc[6].

- Step 1: Build the original data matrix

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

- Step 2: Standardize the metrics to eliminate the effect of dimensionality

$$x_{ij} = \frac{a_{ij} - \min a_{ij}}{\max a_{ij} - \min a_{ij}}$$

$$x_{ij} = \frac{\max a_{ij} - a_{ij}}{\max a_{ij} - \min a_{ij}}$$

Where, x_{ij} is the standardized index value, $\max a_{ij}$ and $\min a_{ij}$ are the maximum and minimum values of the j th index, respectively. Since the standardized results need to do logarithmic processing, there cannot be a value of 0, so the standardized data will be added 0.0001 each, that is, $p_{ij} = x_{ij} + 0.0001$

- Step 3: Calculate the information entropy, m is the number of countries

$$e_j = -\frac{1}{\ln m} \sum_{i=1}^m \frac{p_{ij}}{\sum_{i=1}^m p_{ij}} \ln \frac{p_{ij}}{\sum_{i=1}^m p_{ij}}$$

- Step 4: Calculate the weights, w_j is the weight

$$w_j = \frac{1 - e_j}{\sum_{j=1}^n 1 - e_j}$$

- Step 5: Calculate the weighted distance from the maximum and minimum values

$$D_i^+ = \sum_{j=1}^m w_j (\max p_{ij} - p_{ij})^2$$

$$D_i^- = \sum_{j=1}^m w_j (\min p_{ij} - p_{ij})^2$$

- Step 6: Calculate the final score

$$S_i = \frac{D_i^-}{D_i^+ + D_i^-}$$

4.1.2 Indicator System Construction

Table2 : Indicator System and Attribute

Level 1	Level 2	Level 3	Unit	Attribute	Correlation	Code
Social stability level	Resource endowment	Population density	People/km2	1	Negative	A1
		Freshwater per capita	Cubic meter/person	1	Positive	A2
	Economic situation	GDP per capita	USD	1	Positive	B1
		Gini coefficient	%	1	Negative	B2
	Medical level	Hospital beds per capita	Sheet/thousand people	1	positive	C1
		Doctors per capita	Person/thousand	1	Positive	C2
	Education level	Literacy rate	%	1	Positive	D1
		Education Expenditure	%	1	Positive	D2
	Technology level	RD spending	%	1	positive	E1
		Patent per capita	Person/thousand	1	Positive	E2
		High and New Technology Products	%	1	Positive	E3
	Personal pressure	Dependency Ratio	%	1	Negative	F1
		Unemployment rate	%	1	Negative	F2
	Comprehensive Strength	Coastal	1	0	Negative	G1
		Degree of development	1	0	Negative	G2
	Ideology	Social Nature	1	0	Negative	H1
		Religious beliefs	1	0	Negative	H2

In the selection of indicators, we strictly follow the principles of scientific, systematic, dynamic and operable, follow the scientific development law of things, ensure a clear and concise logical relationship between indicators, grasp the objective law that things are constantly changing and developing, and at the same time take into account our own ability to collect and process data.

Finally, we selected 8 secondary indicators and 17 tertiary indicators from the World Bank, the International Monetary Fund, the United Nations website and the China International Statistical Yearbook. In total, the indicators include the country's basic conditions such as resources and population, social security conditions such as education and medical care, and macro indicators such as politics and science and technology of the country. And the qualitative and quantitative indicators are comprehensively analyzed and qualitative indicators are quantified.

4.1.3 Descriptive Statistics

Table 3 shows the results of descriptive statistics for each 3-level indicator, which shows that there are large differences in population density, fresh water per capita, GDP per capita, hospital beds per capita, R&D expenditure, patents per capita, high-tech products, and unemployment rate, indicating that the current international economic and scientific development is uneven and the distribution of resources and population is unbalanced. Meanwhile, the Gini coefficient, literacy rate, and dependency ratio indicators differ less among countries and are more concentrated in distribution. It indicates that with the continuous development of society, education, and the selfless contribution of public welfare organizations, the illiteracy ratio of similar countries is

decreasing again. With the continuous development of industrial society and the increasing life expectancy of population, all countries are facing the aging problem of different severity.

Table3 : Descriptive Statistics of Data

Variable	Obs	Mean	Std. Dev.	Min	Max
A1	80	632.781	2583.696	1.9	19466.4
A2	80	1.037	1.702	0.001	8.019
B1	80	19134.35	19559.38	1084	63544
B2	80	0.394	0.044	0.32	0.63
C1	80	2.87	3.026	0.1	13.7
C2	80	1.514	1.326	0.2	4.6
D1	80	73.07	18.145	56.8	99.8
D2	80	3.82	1.751	0.8	6.7
E1	80	1.053	1.065	0.1	4.3
E2	80	2.468	6.06	0	33.14
E3	80	15.045	14.4	0.3	64.4
F1	80	51.949	9.564	34.5	89.2
F2	80	4.415	4.906	0.1	24.9
G1	80	0.075	0.265	0	1
G2	80	0.025	0.157	0	1
H1	80	0.05	0.219	0	1
H2	80	0.775	0.656	0	2

4.1.4 Weighting Analysis

First, Figure 3 shows the bar graphs of the percentages of eight 2-level indicators in 2014 and 2020, which can show the change of indicator weights in 2014 and 2020. In general, in the social stability risk assessment model, except for the significant increase in the proportion of pressure on medical level, the proportion of education level and personal pressure slightly decreases, and the weight of the remaining 2-level indicators decreases, indicating that with the development of modern science and technology and communication technology, residents of all countries pay more attention to medical level, while residents pay more attention to personal development including education and personal pressure, and the overall macro situation of the country. On the contrary, the influence on social stability is reduced, indicating that the residents of the current society are more concerned about their living environment and pursue higher personal development.

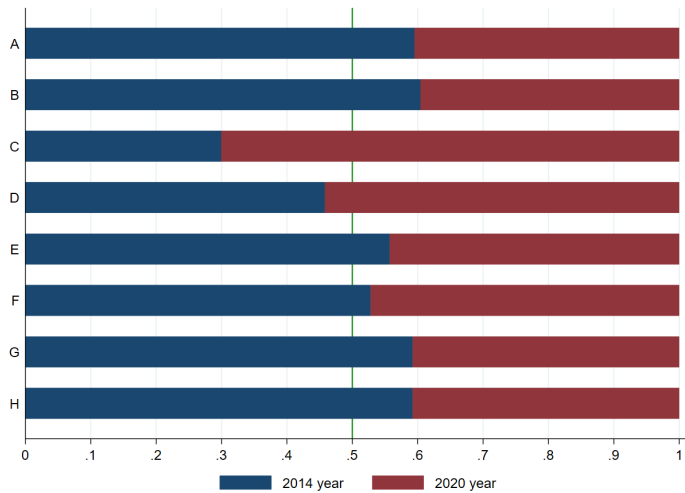


Figure3 : Two-level indicator percentage bar chart

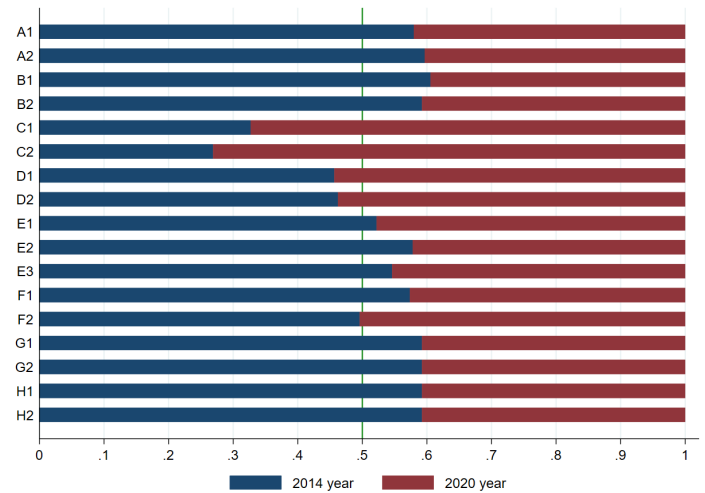


Figure4 : Three-level indicator percentage bar chart

Second, Figure 4 shows a bar chart of the percentage of level 3 indicators, with small overall changes and decreasing weights for most indicators, indicating that the social stability situation in 2020 is more complex than that in 2014. Among them, the medical level contains two indicators that significantly increase, which also confirms the conclusion of Table 3. The rest of the indicators changed less.

Third, Figure 5 shows the heat map of the 3-level indicator weights in 2014, which shows that the residents of the sample countries pay extra attention to freshwater resources per capita, GDP per capita, literacy rate, RD expenditure, and patents per capita, indicating that residents pay higher attention to scientific and technological development as well as higher quality development of individuals.

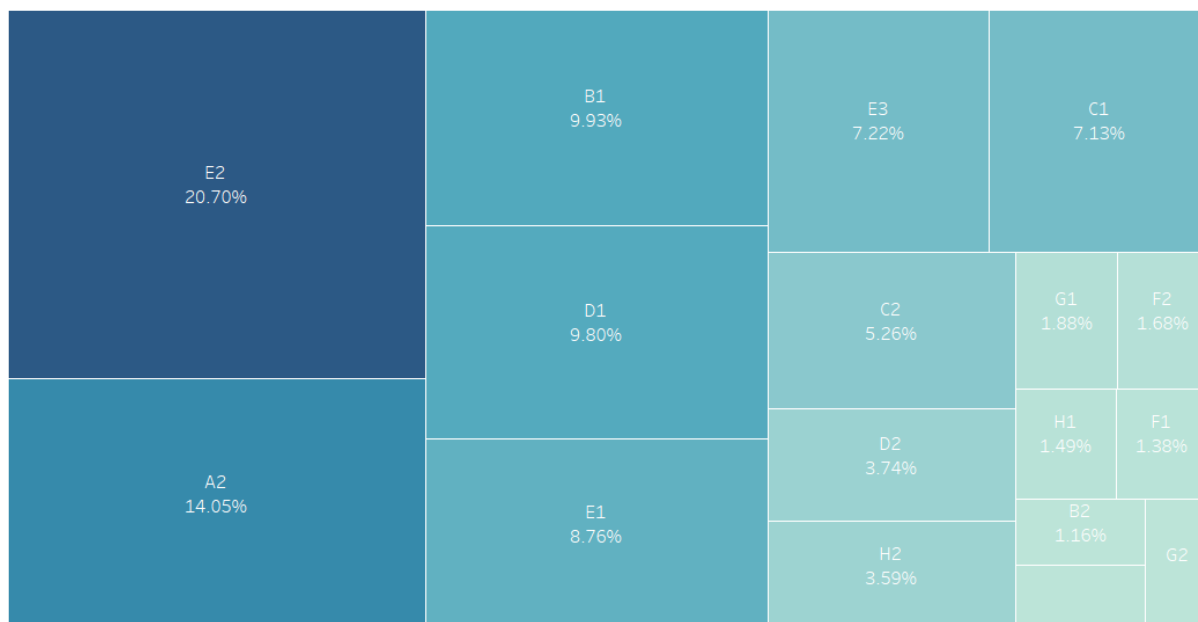


Figure5 : Heat map of metrics in 2014

Finally, Figure 6 shows a heat map of the 3-level indicators for 2020, which shows that the top 5 weightings are fresh water per capita, hospital beds per capita, doctors per capita, literacy rate, and patents per capita. The most striking feature is the increased focus on personal health compared to 2014.

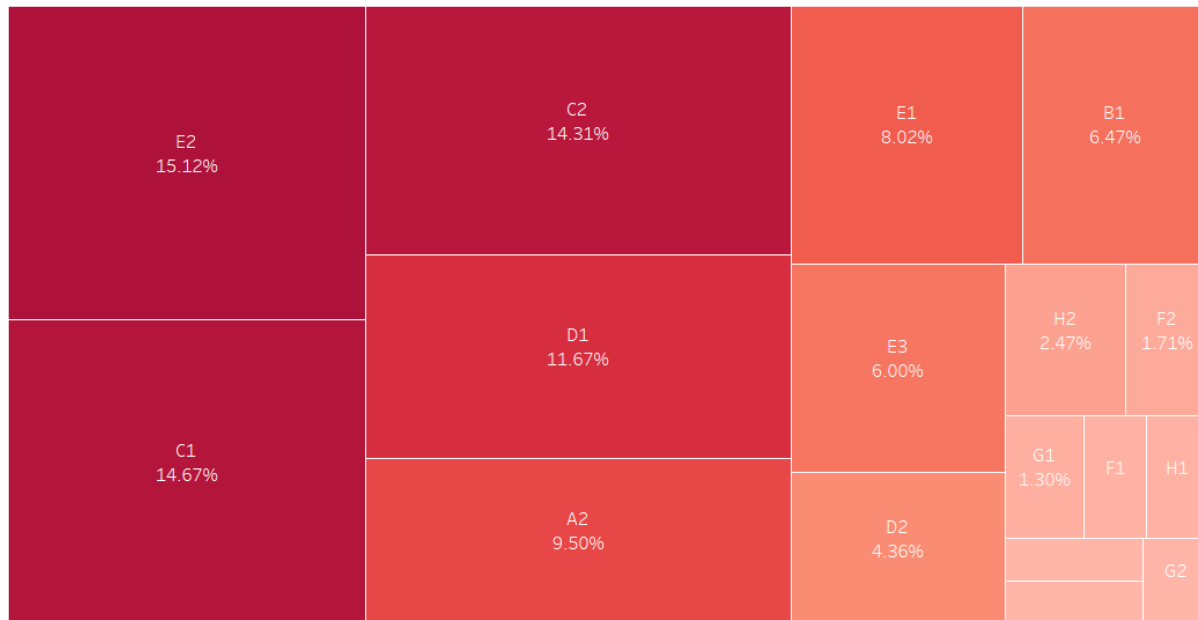


Figure6 : Heat map of metrics in 2020

4.1.5 Distribution of Entropy-Weighted TOPISIS Scores

Figure 7 shows the kernel density plot of the entropy-weighted-TOPISIS scores, which shows that compared to 2014 for each indicator of social stability, the scores of countries in general have improved in 2020, with more countries having a higher composite stability index and the emergence of several countries that are more stable than in 2014, but there are still a few countries that have the same scores as the lowest countries in 2014.

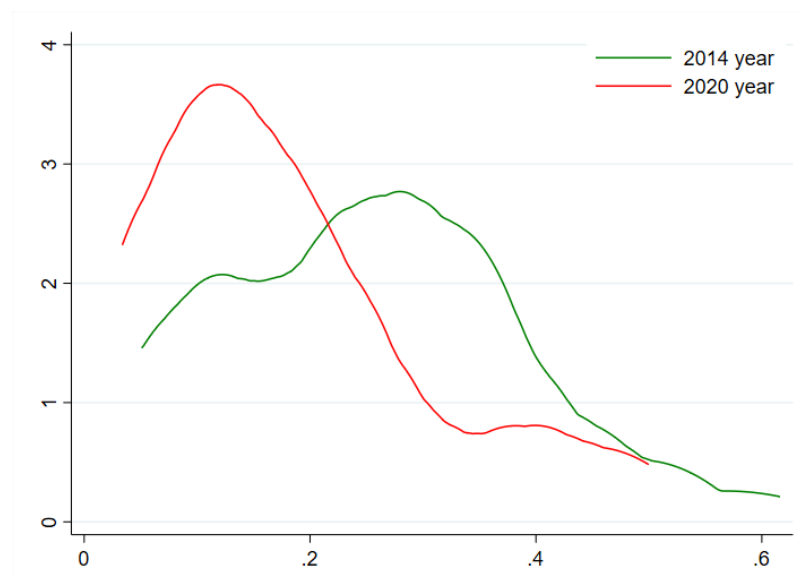


Figure7 : Kernel density plot of entropy-weighted TOPISIS scores

4.1.6 Correlation Analysis Between Indicators

The Spearman correlation coefficient, or spearman correlation coefficient, has no limitations of continuous data, normal distribution, and linear relationship relative to the person coefficient. Therefore, to obtain more accurate results, we use Spearman's correlation coefficient. It is a nonparametric measure of the dependence of two variables, and it evaluates the correlation of two statistical variables using monotonic equations. The Spearman's correlation coefficient is +1 or -1 if there are no repeated values in the data and when the two variables are perfectly monotonically correlated.

Figure 8 shows the Spearman correlation coefficient plot for 2014, from which it can be seen that overall there are fewer negative correlations and weaker correlations among the indicators, and those with strong correlations are basically positive, with GDP per capita having strong positive correlations with hospital beds per capita, doctors per capita, R&D expenditure, patents per capita, and high-tech products, and hospital beds per capita having strong positive correlations with doctors per capita, R&D expenditure, patents per capita patents, high-tech products have strong positive correlations, and R&D expenditure, patents per capita, high-tech products have strong correlations with each other, indicating that there is a close relationship between the level of economic development, medical medicine and the level of scientific and technological development.

V	A1	A2	B1	B2	C1	C2	D1	D2	E1	E2	E3	F1	F2	G1	G2	H1	H2
A1	1	-0.69	-0.08	0.151	-0.01	-0.14	-0.1	-0.19	0.043	-0.02	0.235	-0.01	-0.17	0.045	-0.1	-0.07	-0.09
A2	-0.69	1	0.007	-0.26	-0.02	-0.04	-0.07	0.149	-0.18	-0.05	0.009	-0.14	-0.22	0.127	-0.08	0.06	0.127
B1	-0.08	0.007	1	0.214	0.474	0.521	-0.08	0.306	0.664	0.594	0.488	-0.17	0.297	-0.21	-0.06	-0.06	-0.62
B2	0.151	-0.26	0.214	1	-0.19	0.021	0.427	0.021	-0.11	-0.08	0.061	-0.1	0.244	-0.11	0.044	-0.16	0.073
C1	-0.01	-0.02	0.474	-0.19	1	0.66	0	0.387	0.562	0.586	0.436	-0.13	0.265	-0.02	-0.27	0.08	-0.53
C2	-0.14	-0.04	0.521	0.021	0.66	1	0.162	0.542	0.702	0.681	0.328	-0.09	0.645	-0.22	-0.06	-0.04	-0.71
D1	-0.1	-0.07	-0.08	0.427	0	0.162	1	-0.05	-0.14	-0.05	-0.15	-0.36	0.208	-0.13	0.007	-0.25	0.202
D2	-0.19	0.149	0.306	0.021	0.387	0.542	-0.05	1	0.443	0.346	0.415	-0.07	0.467	-0.01	-0.13	-0.02	-0.45
E1	0.043	-0.18	0.664	-0.11	0.562	0.702	-0.14	0.443	1	0.828	0.569	-0.11	0.451	-0.12	-0.11	-0.04	-0.81
E2	-0.02	-0.05	0.594	-0.08	0.586	0.681	-0.05	0.346	0.828	1	0.545	-0.28	0.434	-0.09	0.146	-0.11	-0.8
E3	0.235	0.009	0.488	0.061	0.436	0.328	-0.15	0.415	0.569	0.545	1	-0.28	0.052	0.107	-0.23	-0.13	-0.56
F1	-0.01	-0.14	-0.17	-0.1	-0.13	-0.09	-0.36	-0.07	-0.11	-0.28	-0.28	1	0.014	-0.13	-0.19	0.119	-0.02
F2	-0.17	-0.22	0.297	0.244	0.265	0.645	0.208	0.467	0.451	0.434	0.052	0.014	1	-0.26	0.215	-0.11	-0.5
G1	0.045	0.127	-0.21	-0.11	-0.02	-0.22	-0.13	-0.01	-0.12	-0.09	0.107	-0.13	-0.26	1	-0.05	0.37	0.242
G2	-0.1	-0.08	-0.06	0.044	-0.27	-0.06	0.007	-0.13	-0.11	0.146	-0.23	-0.19	0.215	-0.05	1	-0.04	0.069
H1	-0.07	0.06	-0.06	-0.16	0.08	-0.04	-0.25	-0.02	-0.04	-0.11	-0.13	0.119	-0.11	0.37	-0.04	1	0.243
H2	-0.09	0.127	-0.62	0.073	-0.53	-0.71	0.202	-0.45	-0.81	-0.8	-0.56	-0.02	-0.5	0.242	0.069	0.243	1

Figure8 : Spearman's correlation coefficient of the tertiary indicator for 2014

Figure 9 shows the Spearman correlation coefficient plot for 2020, and the significant changes that can be found relative to 2014 are an increase in the correlation between GDP per capita and education expenditure, a weakening of the correlation with the number of doctors per capita, and a weakening of the correlation between doctors per capita and the number of beds per capita, indicating that as the level of economic development increases, people are more likely to receive higher education and are willing to pay more for it, and that the core constraint on the development of healthcare in a region is the number of doctors, and the number of beds does not have a significant effect.

V	A1	A2	B1	B2	C1	C2	D1	D2	E1	E2	E3	F1	F2	G1	G2	H1	H2
A1	1	-0.11	-0.2	0.31	-0.1	-0.08	0.232	-0.22	-0.19	-0.23	-0.12	0.063	-0.11	-0.23	-0.13	-0.19	0.141
A2	-0.11	1	0.01	-0.25	-0.05	-0.17	-0.14	0.004	-0.15	-0.09	-0.11	-0.15	-0.18	0.119	-0.08	0.04	0.136
B1	-0.2	0.01	1	-0.01	0.514	-0	-0.45	0.435	0.792	0.808	0.565	0.068	0.162	-0.16	-0.17	-0.07	-0.87
B2	0.31	-0.25	-0.01	1	-0.21	-0.07	0.271	0.058	-0.07	-0.08	-0.02	-0.13	-0.1	-0.11	0.044	-0.16	0.073
C1	-0.1	-0.05	0.514	-0.21	1	0.212	-0.32	0.369	0.56	0.57	0.296	0.291	0.107	-0.19	-0.11	0.119	-0.54
C2	-0.08	-0.17	-0	-0.07	0.212	1	0.007	0.203	0.014	0.109	-0.02	0.191	0.238	-0.2	0.196	0.129	-0.11
D1	0.232	-0.14	-0.45	0.271	-0.32	0.007	1	-0.21	-0.39	-0.39	-0.21	-0.13	-0.25	0.164	0.171	0.332	0.485
D2	-0.22	0.004	0.435	0.058	0.369	0.203	-0.21	1	0.341	0.358	0.256	-0	-0.13	-0.25	-0.02	-0.17	-0.42
E1	-0.19	-0.15	0.792	-0.07	0.56	0.014	-0.39	0.341	1	0.831	0.585	0.087	0.282	-0.16	-0.16	-0.03	-0.75
E2	-0.23	-0.09	0.808	-0.08	0.57	0.109	-0.39	0.358	0.831	1	0.507	-0.05	0.285	-0.07	0.118	-0.11	-0.78
E3	-0.12	-0.11	0.565	-0.02	0.296	-0.02	-0.21	0.256	0.585	0.507	1	-0.15	0.009	0.111	-0.24	-0.09	-0.54
F1	0.063	-0.15	0.068	-0.13	0.291	0.191	-0.13	-0	0.087	-0.05	-0.15	1	0.124	-0.14	-0.16	0.219	-0.21
F2	-0.11	-0.18	0.162	-0.1	0.107	0.238	-0.25	-0.13	0.282	0.285	0.009	0.124	1	-0.05	0.281	-0.05	-0.24
G1	-0.23	0.119	-0.16	-0.11	-0.19	-0.2	0.164	-0.25	-0.16	-0.07	0.111	-0.14	-0.05	1	-0.05	0.37	0.242
G2	-0.13	-0.08	-0.17	0.044	-0.11	0.196	0.171	-0.02	-0.16	0.118	-0.24	-0.16	0.281	-0.05	1	-0.04	0.069
H1	-0.19	0.04	-0.07	-0.16	0.119	0.129	0.332	-0.17	-0.03	-0.11	-0.09	0.219	-0.05	0.37	-0.04	1	0.243
H2	0.141	0.136	-0.87	0.073	-0.54	-0.11	0.485	-0.42	-0.75	-0.78	-0.54	-0.21	-0.24	0.242	0.069	0.243	1

Figure9 : Spearman's correlation coefficient of the tertiary indicator for 2020

4.1.7 Cause and Effect Analysis Between Indicators

➤ GDP per capita and number of hospital beds per capita

In the materialistic view of history, the economic base determines the superstructure, productivity determines the relations of production, and productivity is the most dynamic and central factor in social change. Based on the previous study, we further explore the causal relationship between GDP per capita and the number of hospital beds per capita. For GDP per capita and the number of hospital beds per capita, it is obvious to find that it is the increase in GDP per capita, the increase in the level of socio-economic development, the increase in the wealth that people can spend on medical care, and therefore the increase in the number of hospital beds.

➤ GDP per capita and education spending

Building on the research in the previous question, we continue to explore the causal relationship between GDP per capita and education spending. In line with the logic of the previous question, GDP per capita and education expenditure are also the result of higher GDP per capita as a cause and higher education expenditure as a consequence. As the social economy continues to grow and social wealth increases, people have more disposable income, and then the education expenditure spent on the next generation will also increase accordingly.

➤ GDP per capita and R&D expenditure

Similar to the previous question, regarding the causal relationship between GDP per capita and R&D expenditure, the conclusion is that an increase in GDP per capita leads to an increase in R&D expenditure, because only when the level of economic development increases, the government's fiscal revenue will increase and more fiscal expenditure will be available to invest in science and technology development and promote innovation and research in science and technology.

➤ GDP per capita and high-tech products

As for high-tech products and GDP per capita, this paper believes that it is the continuous emergence of high-tech products that promotes the competitiveness of the country's export products, as well as the improvement of production processes, production technology, and production efficiency, which promotes the country's exports as well as domestic residents' consumption and raises GDP per capita.

➤ R&D expenditure and patents per capita

Regarding the relationship between R&D expenditure and patents per capita, we can safely assume that the increase in R&D expenditure leads to the development of science and technology and the increase of scientific and technological talents, which further leads to the innovation and development of scientific research, creating more patents and increasing the number of patents per capita.

➤ Patents per capita and high-tech products

Regarding the relationship between patents per capita and high-tech products, we believe that it is the increase in the number of patents per capita that leads to an increase in a region's productivity level, improved production processes, enhanced production efficiency, which promotes the development of production technology and increases the proportion of high-tech products in the region.

4.2 Task 2

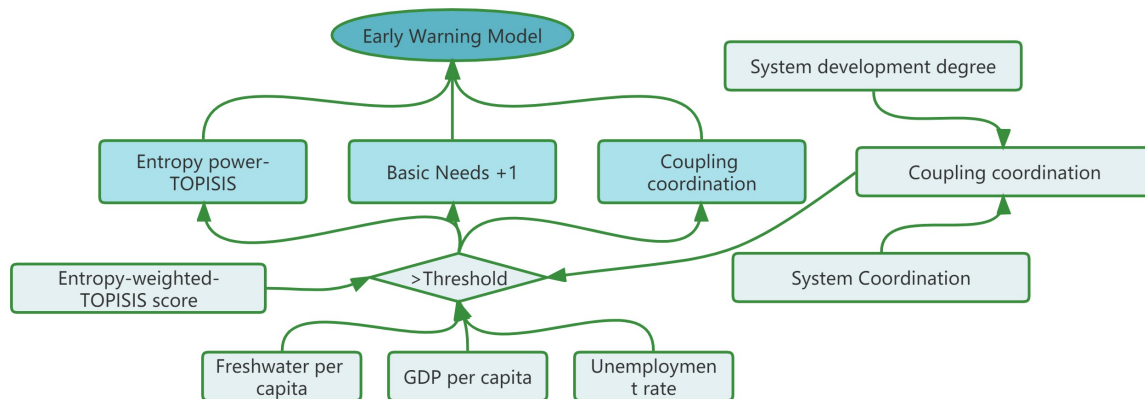


Figure10 : Work flow chart

4.2.1 Coupling Coordination Degree Model Construction

Society is a complex system with multiple intrinsic coupling relationships among different systems such as resources, ecology, economy and society. With the deepening understanding of the scientific concept of development, the judgment of the development degree of a region or society has shifted from the pure development level to the evaluation of the overall balanced development based on the coordination level and development level. Therefore, coupling effect and coupling coordination development degree have become effective evaluation research tools.[7] Among them, the coupling coordination degree model uses the coupling degree to explain the interrelationship between several subsystems, and further uses the coordination development degree to conduct a comprehensive evaluation and research on the whole system. The following are the steps of its construction.

- Step 1: Construct the index matrix

$$[U_1 \ U_2 \ \cdots \ U_{n-1} \ U_n]$$

U_i denotes the standardized value of the i th indicator, and the standardization method is consistent with the previous question.

- Step 2: Calculate the coupling degree

$$C = \sqrt[n]{\frac{\prod U_i}{(\sum U_i)^n}}$$

C denotes the coupling degree between n indicators

- Step 3: Calculate the comprehensive evaluation index

$$T = \sum \omega_i * U_i$$

T denotes the comprehensive development level of n indicators, and ω_i denotes the weight of this sub-indicator

- Step 4: Calculate the coupling coordination degree

$$D = \sqrt[2]{T * C}$$

D denotes the coupling coordination degree of n indicators

4.2.2 Early Warning Model Construction

Table4 : Threshold setting situation

Indicator	Attribute	Correlation	Unit	Threshold
A2	Basic Needs	Positive	Cubic meter/person	-
B1	Basic Needs	Positive	USD	-
F2	Basic Needs	Negative	%	-
D	Coupling coordination degree	Positive	1	-
AS	Entropy-weighted-TOPISIS score	Positive	1	-

A stable society requires comprehensive coordination of all aspects. In order to avoid a model in which the final score of a society with obvious disadvantages is pulled up by its advantages, which leads to the neglect of disadvantages. As shown in Table 4, we developed an early warning model for social stability. We set the threshold by calculating the 3 basic social needs, the final score and the coupling coordination, and if the coordination is lower than the base threshold, it means that there is an imbalance in the development of its society, which is in an unhealthy state and remains a greater risk. Accordingly, we set up an early warning model of social stability.

Table 5 shows our social stability early warning model, with 3 indicators set as composite score, coupling coordination degree, and 3 basic needs scores. The full scores are 1, 1, and 3. By dividing the scores of the 3 indicators, we set 5 risk levels from no risk to mild risk, moderate risk, near severe, and severe risk.

Table5 : Early warning level table

AS	D	Specific indicators	Early Warning Level
0	0	0	No risk
0	1	0	Mild risk
0	0	1	Light risk
1	0	0	Moderate risk
0	1	1	Moderate risk
0	0	2 OR 3	Moderate risk
0	1	2 OR 3	Near Severe
1	1	0	Near Severe
1	0	1	Near severe
1	0	2 OR 3	Severe Risk
1	1	1	Severe Risk
1	1	2 OR 3	Very serious risk

4.3 An Analysis of the Color Revolution in Cambodia

4.3.1 Descriptive statistics of assessment indicators

Table6 : Descriptive statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
A2	40	1.066	1.775	0.002	8.019
B1	40	18734.53	19707.7	1084	61887
F2	40	5.845	5.574	0.1	24.9
D	40	0.389	0.139	0.074	0.579
AS	40	0.25	0.136	0.051	0.616

Descriptive statistics of the three basic demand indicators, coupling coordination and final scores are shown in Table 6. It can be seen that the distribution of all 5 indicators is relatively unbalanced with large extreme differences, reflecting the huge differences in the current international distribution of resources, economic level, individual social pressure and social stability risks.

4.3.2 Threshold Criteria and Cambodia Score

Table7 : Threshold criteria and Cambodia score

Indicator	Attribute	Threshold	Cambodia	Result
A2	Basic Needs	0.7	0.7827	0
B1	Basic Needs	1200	1084	1
F2	Basic Needs	0.2	0.1	0
D	Coupling coordination degree	0.2	0.14337	1
AS	Entropy-weighted-TOPISIS score	0.12	0.052776	1

Since there was a color revolution in Cambodia that tried to overthrow the regime, we selected Cambodia as the subject of this phase of the study. Table 7 shows the social early warning model after setting the threshold criteria and Cambodia's score in this model. From the above descriptive statistics results, we can see that among

the three basic needs of Cambodia, the level of economic development is slightly below the threshold and the unemployment rate is low, but the coupling coordination is poor, indicating the imbalance of local social development and the final score is low. According to the results of our risk warning model, Cambodia is in the tier of severe risk.

After normalizing the above five indicators according to their positivity and negativity, their kernel density plots are drawn, and the results are shown in Figure 11. It can be found that, relatively speaking, each country performs well in the unemployment rate, but performs poorly in the freshwater resources per capita and needs to further improve environmental protection and conserve freshwater resources, and the overall score, coupling coordination and GDP per capita distribution are more balanced.

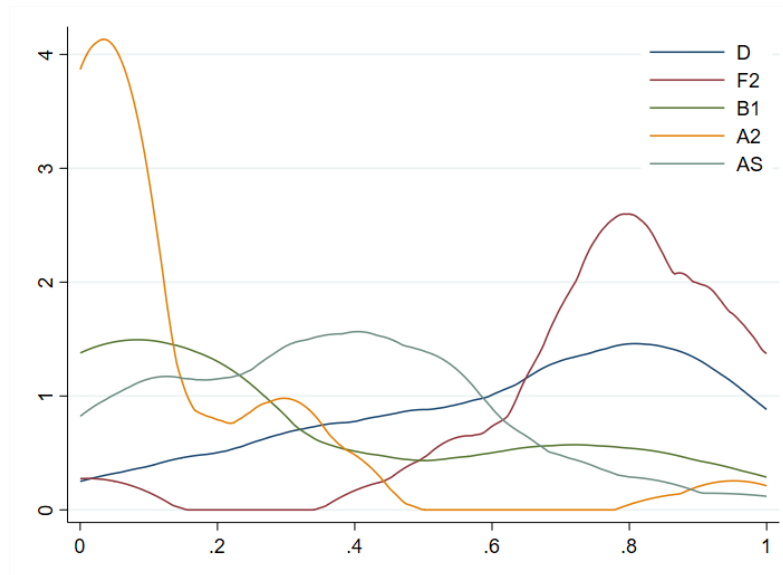


Figure11 : Kernel density plot of Cambodia score

4.3.3 Analysis of Reasons for Failure

➤ Religious Beliefs

In terms of religious beliefs, Buddhist ideas are widely spread in Cambodia, the country is relatively unified, and Buddhist ideology contributes to social stability and harmony, and throughout Cambodia's history, authority has often been combined with Buddhist ideas to form a more solid political form.

➤ Dependency Ratio

The dependency ratio is low and there is less pressure on young people in society. Cambodia's low dependency ratio indicates that Cambodian residents face less pressure to raise, while ensuring the quality of life of the elderly, contributing to the general well-being of the society and, to a certain extent, freeing up a large number of young people to engage in value-creating activities, thus enhancing the vitality of economic development.

➤ Freshwater per capita

The abundance of fresh water per capita gives the local community an advantage in terms of resources and the ability to deal with population and resource conflicts in an appropriate manner, preventing problems arising from insufficient fresh water resources, and allowing sufficient fresh water resources to be used for industrial mass production, promoting economic development and enhancing social stability and harmony.

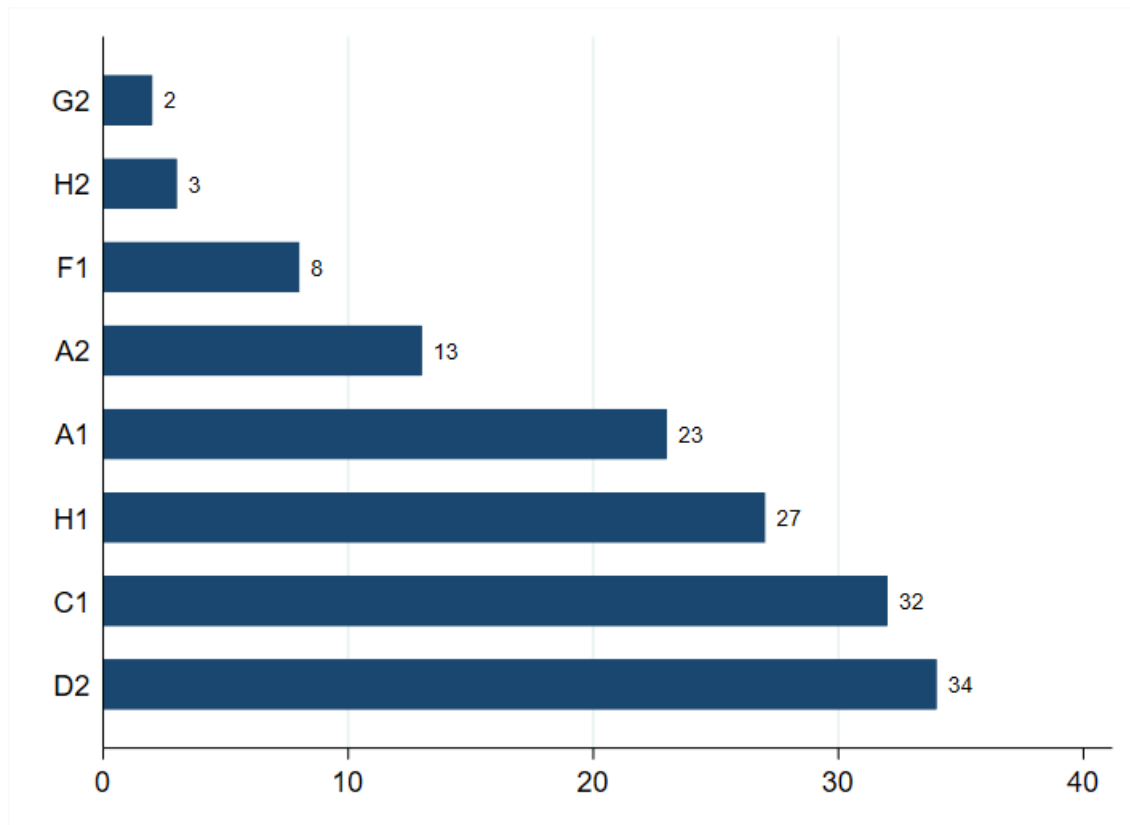


Figure12 : The 8 indicators with top scores in Cambodia

4.3.4 Future Projections

The future forecast for Cambodia concludes that the probability of a color revolution in Cambodia in the future is low.

First, social development is a complex and long process, unless there is a technological explosion or a large-scale war in the future, a society will continue its inherent development pattern and develop slowly.

Secondly, Cambodia is still one of the underdeveloped countries with a low level of economic and technological development, and will continue to develop in the future with a low level of comprehensive development and social stability.

Thirdly, Cambodia is a constitutional monarchy and has been colonized by Western countries for a long time. This colonial history has made the Cambodian people hate external forces, so they are more likely to consider the problems from their own perspective when they are provoked by foreign forces.

Finally, Cambodia is located in the south-central peninsula, with plains, mountains, and plateaus in the country, and is bordered by China in the southwest. The probability of a color revolution is low.

4.4 An Analysis of the Color Revolution in Myanmar

4.4.1 Descriptive Statistics of Assessment Indicators

Table8 : Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
A2	40	1.009	1.647	0.001	7.499
B1	40	19534.18	19652.65	1194	63544
F2	40	2.985	3.673	0.3	11.7
D	40	0.311	0.142	0.06	0.554
all score	40	0.178	0.122	0.034	0.5

We selected the successful example of color revolution, Myanmar, as our research object, and Table 8 shows its descriptive statistics. It can be seen that the extreme differences of freshwater per capita, GDP per capita and unemployment rate are all large, and the overall score distribution varies widely.

After normalizing the above five indicators according to their positivity and negativity, their kernel density plots are drawn, and the results are shown in Figure 13. It can be found that, relatively speaking, freshwater per capita performs poorly and needs to further improve environmental protection and conserve freshwater resources, while the recession of unemployment rate, which performed better in 2014, instead, indicates that the new crown epidemic in 2020 has a significant contribution to the rise of unemployment rate. The composite score, coupling coordination and GDP per capita are consistent with 2014, with a more balanced distribution.

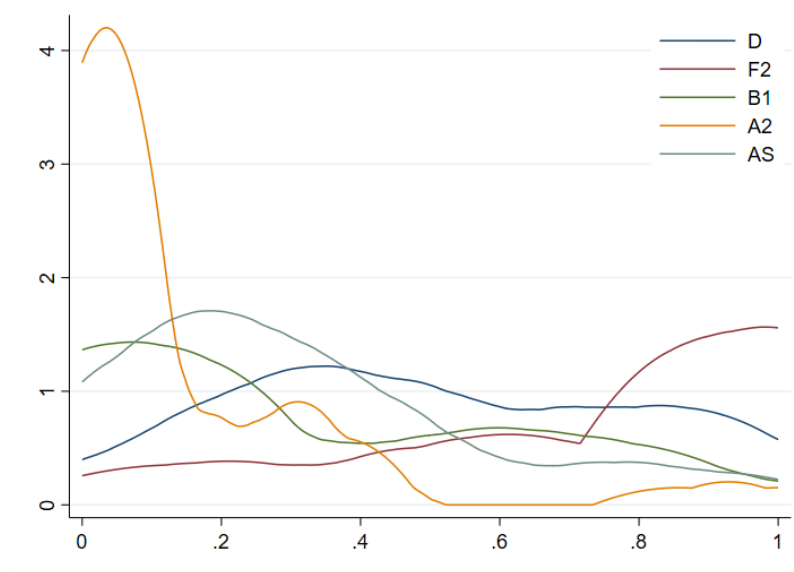


Figure13 : Kernel density map of Myanmar score

4.4.2 Threshold Criteria and Cambodia Score

Table9 : Threshold criteria and Myanmar score

Indicator	Attribute	Threshold	Cambodia	Result
A2	Basic Needs	0.7	1.843411	0
B1	Basic Needs	1500	1400	1
F2	Basic Needs	0.2	0.3	1
D	Coupling coordination degree	0.3	0.29367	1
AS	Entropy-weighted-TOPISIS score	0.15	0.1298	1

Subsequently, as shown in Table 9, we set the threshold criteria and calculate the score for Myanmar, where the unemployment rate is high, 50% above the threshold, and the GDP per capita is low, and the final result shows that the social stability risk in Myanmar is at a severe risk level.

4.4.3 Analysis of the Reasons for Success

➤ Unemployment rate

With high unemployment rate, young people face greater pressure of unemployment, which leads to lower social well-being. At the same time, a large number of unemployed people will also impact social stability, which is not conducive to social harmony and stability. The lower unemployment rate also reflects certain misunderstandings in local government policies and the downturn of the local economy, which are risks to social stability.

➤ R&D expenditure

R&D spending is low. Science and technology is the cornerstone of economic development, and according to our research, R&D spending can be effectively transformed into high-tech products and patents to promote the development of production technology, which in turn promotes economic development. Myanmar government's low R&D investment and low dynamics of economic growth make it difficult to sustain long-term economic development.

➤ GDP per capita

The low GDP per capita reflects the backwardness of Myanmar's economic development, the limited social services the state can provide, and the low standard of living of the people, which makes it difficult to meet their needs and is not conducive to social harmony and stability.

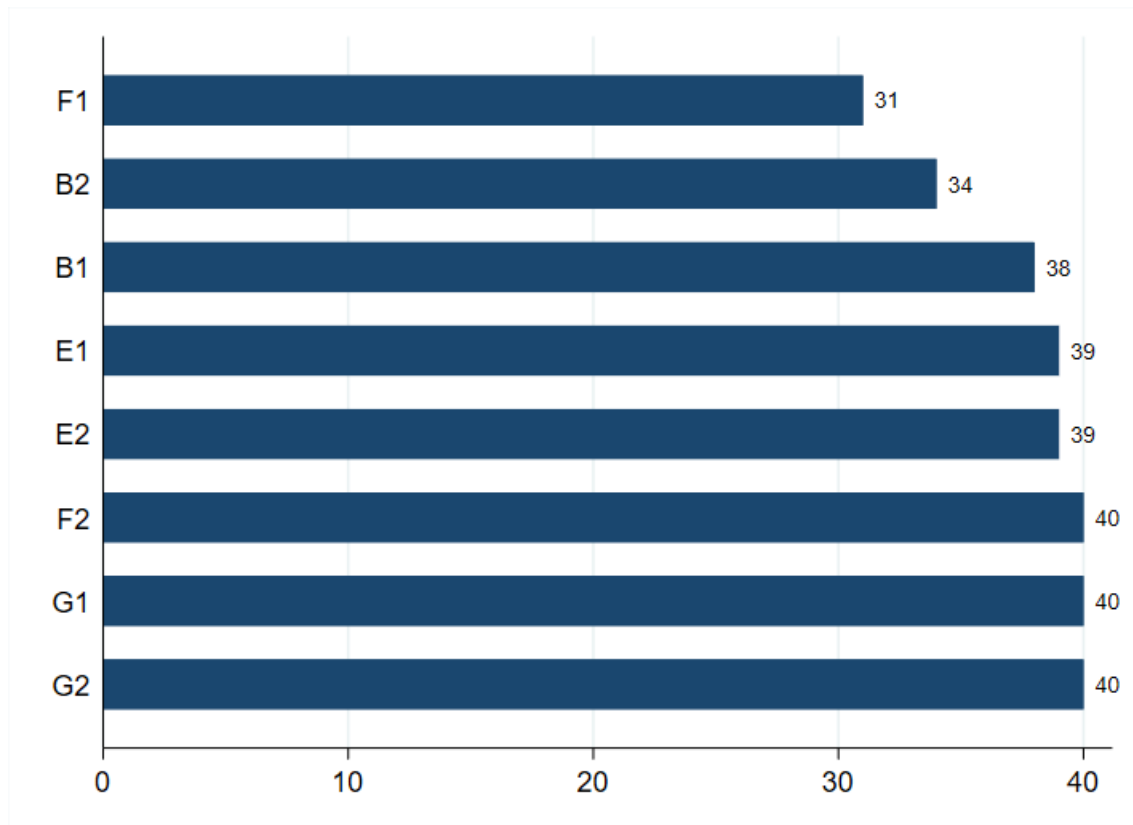


Figure14 : The 8 indicators with lower scores in Myanmar

5 Model Evaluation and Further Discussion

5.1 Strengths

➤ **Indicators are scientific and reasonable**

The construction of social stability index system fully considers qualitative and quantitative indicators, follows scientific, systematic, dynamic and operable, fully and comprehensively covers all aspects of social stability, and makes the evaluation results scientific and reasonable.

➤ **Weights is objective and scientific**

The entropy weight-TOPSIS method cleverly links the objective weighting method with the rigorous multidimensional ranking, which makes the results objective and effectively reflect the real situation, and can help decision makers make scientific evaluation.

➤ **Coordination between subsystems is considered**

The established social stability risk early warning model fully considers the coupling and coordination among indicators to prevent the imbalance of social development from being ignored and fully reflects the interconnection of factors among social systems.

➤ **Applied in practice and the results are good**

The model has been used to analyze two color revolution events, which is in line with the actual results, and the practical application of the model is strong, which can provide some guidance for realistic political decision-making.

5.2 Weaknesses

➤ **Existence of multicollinearity**

Due to the complexity of the real situation, this model simplifies the content of social stability, which leads to the possible existence of multicollinearity and autocorrelation between indicators, and there is a certain error between the results and the actual situation.

➤ **Weights are all objective**

The objectivity of the model is too strong, and the process of calculating the weights of both qualitative and quantitative indicators is using the objective entropy weight method, which may deviate from the actual situation.

5.3 Further Discussion

➤ **Solve the problem of multicollinearity**

Solve the problem of multiple covariance and autocorrelation of model index setting by difference method or data transformation method, which makes the index more scientific and reasonable, and then the calculated result is more accurate and more consistent with reality.

➤ **Consult experts to get more accurate and reasonable weighting.**

For qualitative indicators and some quantitative indicators, the Delphi method can be applied, and the weights can be derived by consulting with experts and integrating the opinions of multiple experts.

6 Advice

➤ Focus on economic development

Ensure a solid increase in the total value of social wealth. Economic development is the cornerstone of social stability, and Maslow's needs theory states that people only have strong expectations for higher-level needs after they have completed their lower-level needs. Thus, the key to maintaining social stability is to keep the economy growing steadily. In addition, smooth economic growth is also helpful to counteract the fluctuations brought by external factors stimulating social stability.

➤ Capitalizing on the Demographic Dividend

It is important to take advantage of the demographic advantage to ease the conflict between population and resources. The comparison between the success and failure of the color revolutions in Cambodia and Myanmar shows that the handling of population is key. In countries with a large population, it is necessary to deal with the contradiction of population and resources, to give full play to the advantages of human resources, and then to obtain the original accumulation of capital, as well as to complete the cultivation and utilization of talents, and to provide capital support and intellectual support for the telling development of science and technology and the economy.

➤ Increase stable employment

Maintain the continuity and stability of employment environment. The result of external factors on social stability stimulation is to a large extent to cause a large number of unemployed people, and the effective way to solve unemployment is to increase and stabilize employment. On the one hand, we can optimize the urban employment environment, improve the relevant laws and regulations, and promote the fair and market-oriented development of the employment market; on the other hand, we should solve the problem of a large number of unemployed people waiting for work in rural areas, and with the continuous promotion of the urbanization process, we must not previously rural areas with a vast number of people.

➤ Improve the competitiveness of science and technology

Increase the investment in science and technology to speed up the speed and quality of science and technology into productivity. Focus on the cultivation and introduction of scientific and technological talents, apply the introduction policy of one-person-one-policy for top talents, through the introduction of a large number of scientific and technological talents, improve the country's scientific and technological competitiveness, shorten the cycle of transforming scientific and technological inventions into real productivity, catch up with the world development trend, and stand on the world development wave.

➤ Establishment of a safety valve mechanism for social stability

The establishment of "up-and-down communication mechanism" and "third-party mediation mechanism", Through the "upward and downward communication mechanism", the voices of the people at the grassroots level are transmitted to the middle and senior levels, as well as to the central leadership, which has a significant effect on the proper resolution of internal conflicts and the satisfaction of people's needs for a better life. The "third-party mediation mechanism" can effectively prevent the parties in conflict from resorting to extreme violence to solve their problems, and through the mediation of the third party, promote the parties in conflict to adopt scientific, reasonable and legal ways to resolve their disputes.

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Appendix

Stata 代码

```
global clist "A2 B1 C1 C2 D1
D2 E1 E2 E3"
```

```
foreach x of global clist{
    bys year:egen minn`x' =
min(`x')
```

```
    bys year:egen maxx`x' =
max(`x')
```

```
    gen standard`x' = (`x' -
minn`x')/(maxx`x' - minn`x') //
```

```
    replace standard`x' =
standard`x'+0.0001
```

```
}
```

```
global dlist "A1 B2 F1 F2 G1
G2 H1 H2"
```

```
foreach x of global dlist{
    bys year:egen minn`x' =
min(`x')
```

```
    bys year:egen maxx`x' =
max(`x')
```

```
    gen standard`x' = (maxx`x' -
`x')/(maxx`x' - minn`x') //
```

```
    replace standard`x' =
standard`x'+0.0001
```

```
}
```

```
global xlist "A1 A2 B1 B2 C1
C2 D1 D2 E1 E2 E3 F1 F2 G1 G2
H1 H2"
```

```
foreach x of global xlist{
    bys year:egen min_standard`x'
= min(standard`x')
```

```
    bys
        year:egen
max_standard`x' = max(standard`x')
}
```

```
foreach x of global xlist {
    bys year:egen summm`x' =
sum(standard`x')
```

```
    gen Y`x' = standard`x' /
```

```
summm`x'
```

```
}
```

```
foreach x of global xlist{
```

```
    gen h`x' = Y`x' * ln(Y`x')
```

```
    bys year:egen he`x' = sum(h`x')
```

```
    gen e`x' = (-1)*he`x'/ln(40)
```

```
    gen d`x' = 1 - e`x'
```

```
}
```

```
gen uselesshe = 0
```

```
*gen q = 0
```

```
foreach x of global xlist{
```

```
    replace uselesshe = d`x' +
uselesshe
```

```
}
```

```
foreach x of global xlist{
```

```
    gen W`x' = d`x' / uselesshe
```

```
    gen S_`x' = W`x'*standard`x'
```

```
*replace q = q+W`x'
```

```
}
```

```
gen d_w_max = 0
```

```
gen d_w_min = 0
```

```
foreach x of global xlist{
```

```
    gen max_w_`x' = W`x' *
(max_standard`x'-standard`x')^2
```

```
    gen min_w_`x' = W`x' *
(standard`x'-min_standard`x')^2
```

```
    replace d_w_max = d_w_max
+ max_w_`x'
```

```
    replace d_w_min = d_w_min
+ min_w_`x'
```

```
}
```

```
gen
all_score =
d_w_min/(d_w_min+d_w_max)
```

```
use result_SQTP.dta,clear
```

```
gen S_A = S_A1+S_A2
```

```
gen S_B = S_B1+S_B2
```

```
gen S_C = S_C1+S_C2
```

```
gen S_D = S_D1+S_D2
```

```
gen S_E = S_E1+S_E2+S_E3
```

```
gen S_F = S_F1+S_F2
```

```
gen S_G = S_G1+S_G2
```

```
gen S_H = S_H1+S_H2
```

```
global xlist "S_A S_B S_C
S_D S_E S_G S_H"
```

```
foreach x of global xlist{
```

```
    bys year:egen minn`x' =
min(`x')
```

```
    bys year:egen maxx`x' =
max(`x')
```

```
    gen standard`x' = (`x' -
minn`x')/(maxx`x' - minn`x') //
```

```
    replace `x' = standard`x'
+0.0001
```

```
    drop minn`x' maxx`x'
standard`x'
```

```
}
```

```
bys year:egen T =
(S_A+S_B+S_C+S_D+S_E+S_F+
S_G+S_H)/8
```

```
bys year:egen C =
(S_A*S_B*S_C*S_D*S_E*S_F*S
_G*S_H/T^8)^(1/8)
```

```
bys year:egen D = (T*C)^0.5
```